EXPT. NO:- 01.
16-07-2026.

BASIC NETWORKING COMMANDS.

AIM:-
To execute basic networking commands in Linux and Windows Operating system.

NETWORKING COMMANDS:-

1. arp-a:-
It displays the ARP cache table. (IP-to-MAC address mappings).

2. hostname:
Displays the computer's hostname.

3. ipconfig /all:-
Shows complete TCP/IP configuration details.

4. nbtstat -a:-
Displays NetBIOS information and troubleshooting details.

5. netstat:
Displays active connections, listening ports, routing tables and interface statistics.
eg:- netstat -r.

6. nslookup:-
It performs DNS queries and displays DNS information.
eg:- nslookup www.google.com

```
Windows PowerShell
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PS C:\Users\DELL> arp -a

Interface: 192.168.137.1 --- 0x3
  Internet Address      Physical Address      Type
  192.168.137.190       12-fb-93-00-a2-03     static
  192.168.137.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  230.0.0.0             01-00-5e-00-00-00     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x10
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  224.0.1.60            01-00-5e-00-01-3c     static
  239.0.0.0             01-00-5e-00-00-00     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 172.16.76.102 --- 0x12
  Internet Address      Physical Address      Type
  172.16.72.1           7c-5a-1c-cf-be-81     dynamic
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  224.0.1.60            01-00-5e-00-01-3c     static
  239.0.0.0             01-00-5e-00-00-00     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
PS C:\Users\DELL> hostname
```

```
PS C:\Users\DELL> hostname
HDC0124162
PS C:\Users\DELL> ipconfig /? all

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]

where
    adapter             Connection name
                       (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
       /setclassid      Modifies the dhcp class id.
       /showclassid6    Displays all the IPv6 DHCP class IDs allowed for adapter.
       /setclassid6     Modifies the IPv6 DHCP class id.

The default is to display only the IP address, subnet mask and
default gateway for each adapter bound to TCP/IP.

For Release and Renew, if no adapter name is specified, then the IP address
leases for all adapters bound to TCP/IP will be released or renewed.

For Setclassid and Setclassid6, if no ClassId is specified, then the ClassId is removed.

Examples:
    > ipconfig                      ... Show information
    > ipconfig /all                 ... Show detailed information
    > ipconfig /renew               ... renew all adapters
    > ipconfig /renew EL*           ... renew any connection that has its
                                        name starting with EL
    > ipconfig /release *Con*       ... release all matching connections,
                                        eg. "Wired Ethernet Connection 1" or
                                            "Wired Ethernet Connection 2"
    > ipconfig /allcompartments     ... Show information about all
                                        compartments
    > ipconfig /allcompartments /all ... Show detailed information about all
                                        compartments
PS C:\Users\DELL>
```

```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.

PS C:\Users\DELL> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:52955        HDC0124162:52956       ESTABLISHED
  TCP    127.0.0.1:52956        HDC0124162:52955       ESTABLISHED
  TCP    127.0.0.1:52957        HDC0124162:52958       ESTABLISHED
  TCP    127.0.0.1:52958        HDC0124162:52957       ESTABLISHED
  TCP    172.16.76.102:55921    hkg12s32-in-f10:https  TIME_WAIT
  TCP    172.16.76.102:56167    icmaaa-ax-in-f10:https  TIME_WAIT
  TCP    172.16.76.102:58804    13.89.179.12:https     ESTABLISHED
  TCP    172.16.76.102:58807    52.123.226.20:https    ESTABLISHED
  TCP    172.16.76.102:58811    lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.76.102:60807    4.213.25.242:https     ESTABLISHED
  TCP    172.16.76.102:61755    pnmaaa-aq-in-f5:https  TIME_WAIT
  TCP    172.16.76.102:64829    cgh01s04-in-f5:https   ESTABLISHED
PS C:\Users\DELL>
```

7. pathping:-

It combines ping and traceroute functionality with statistics.

8. ping:

Packet Internet Groper command is the best way to test connection nodes.

1. #ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. #ping fully qualified domain name.

9. Route:-

Route command is used to show/manipulate the IP routing table. It is to setup static routes to specific host or networks via an interface.

LINUX COMMANDS:-

1) ip:- A versatile command for managing network interfaces, IP addresses and routing tables. It is used to assign or delete IPs, toggle interface status.

* ip address show
* ip address add 192.168.1.254/24 dev enps03.
* ip address del 192.168.1.254/24 dev enps03.
* ip link set eth0 up.
* ip link set eth0 down.
* ip link set eth0 promisc on
* ip route add default via 192.168.1.254 dev eth0
* ip route get 10.10.1.4.

```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

```
PS C:\Users\DELL> ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

2). ipconfig:

A staple tool historically used for configusing and troubleshooting network interfaces. It has largely been replaced by the ip command.

3) mtr:-

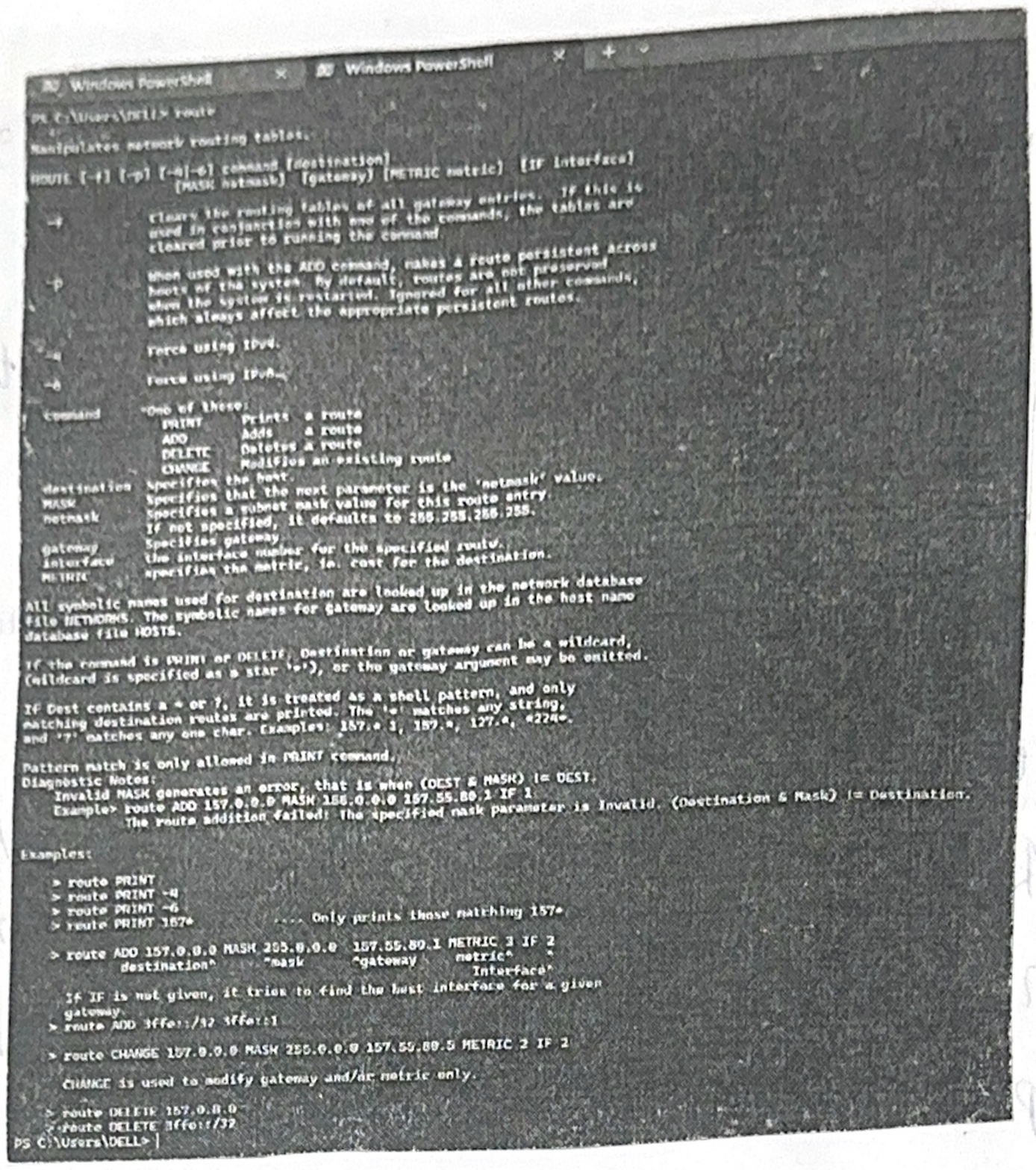
Combines the functionality of ping and traceroute to provide diagnastic statistics for each network hop.

* mtr google.com
* mtr -g google.com.
* mtr -b google.com.
* mtr -C 10 google.com

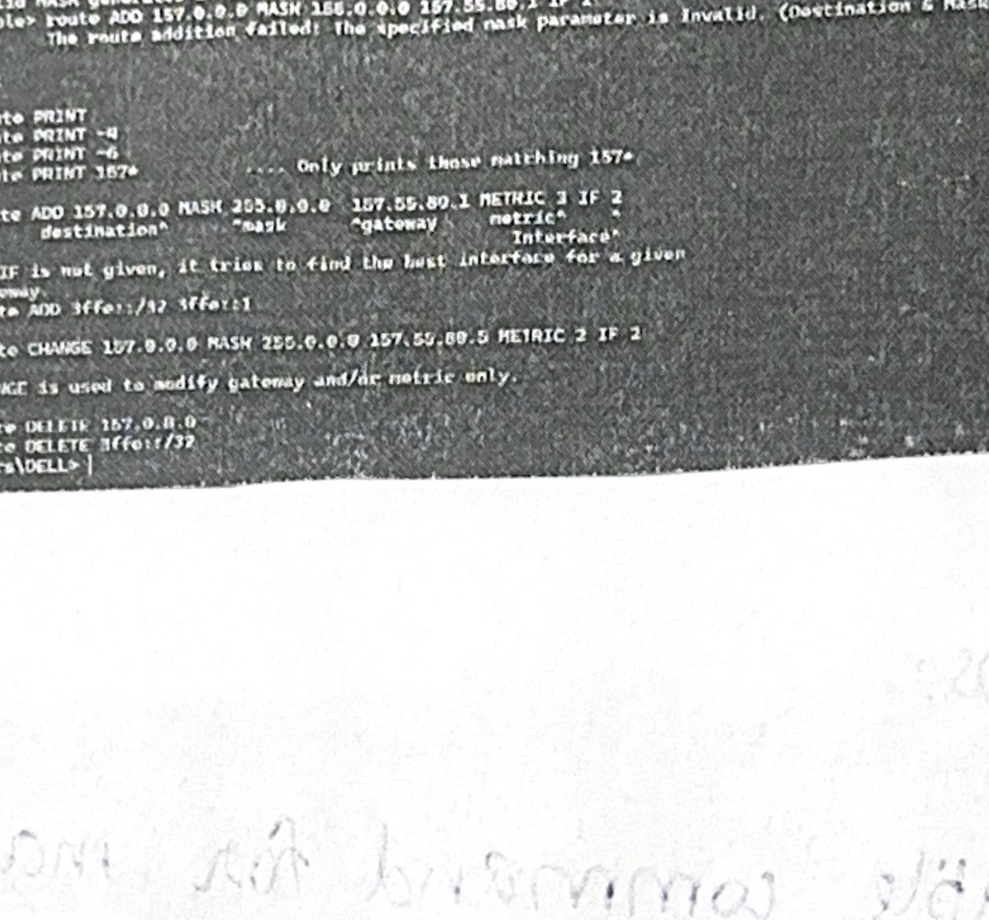
4). tcpdump:-

It is designed for capturing and displaying packets.

* dnf install -y tcpdump.
* tcpdump -D.
* tcpdump -i etb().
* tcpdump -i etb() -C 10
* tcpdump -i etb() -C 10 host 8.8.8.8
* tcpdump -i etb() src host 8.8.8.8.
* tcpdump -i etb() net 10.1.0.0/24
* tcpdump -i etb() port 53.
* tcpdump -i etb() port not 53 and not 25.


Windows PowerShell
Windows PowerShell
PS C:\Users\DELL> route
Manipulates network routing tables.
ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METRIC metric] [IF interface]
-f Clears the routing tables of all gateway entries. If this is used in conjunction with one of the commands, the tables are cleared prior to running the command.
-p When used with the ADD command, makes a route persistent across boots of the system. By default, routes are not preserved when the system is restarted. Ignored for all other commands, which always affect the appropriate persistent routes.
-4 Force using IPv4.
-6 Force using IPv6.
command One of these:
PRINT Prints a route
ADD Adds a route
DELETE Deletes a route
CHANGE Modifies an existing route
destination Specifies the host.
MASK Specifies that the next parameter is the 'netmask' value.
netmask Specifies a subnet mask value for this route entry.
If not specified, it defaults to 255.255.255.255.
gateway Specifies gateway.
interface the interface number for the specified route.
METRIC specifies the metric, ie. cost for the destination.
All symbolic names used for destination are looked up in the network database file NETWORKS. The symbolic names for gateway are looked up in the host name database file HOSTS.
If the command is PRINT or DELETE. Destination or gateway can be a wildcard, (wildcard is specified as a star '*'), or the gateway argument may be omitted.
If Dest contains a * or ?, it is treated as a shell pattern, and only matching destination routes are printed. The '*' matches any string, and '?' matches any one char. Examples: 157.* 1, 157.*, 127.*, *224*.
Pattern match is only allowed in PRINT command.
Diagnostic Notes:
Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.
Examples:
> route PRINT
> route PRINT -4
> route PRINT -6
> route PRINT 157* .... Only prints those matching 157*
> route ADD 157.0.0.0 MASK 255.0.0.0 157.55.80.1 METRIC 3 IF 2
destination^ ^mask ^gateway metric^ ^
Interface^
If IF is not given, it tries to find the best interface for a given gateway.
> route ADD 3ffe::/32 3ffe::1
> route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2
CHANGE is used to modify gateway and/or metric only.
> route DELETE 157.0.0.0
> route DELETE 3ffe::/32
PS C:\Users\DELL>

PS C:\Users\DELL> nslookup
Default Server: UnKnown
Address: 172.16.72.1


MATE Terminal
File Edit View Search Terminal Help
$ ip
Usage: ip [OPTIONS] OBJECT { COMMAND | help }
ip [-force] -batch filename
where OBJECT := { address | addrlabel | amt | fou | help | ila | ioam | l2tp |
link | macsec | maddress | monitor | mptcp | mroute | mrule |
neighbor | neighbour | netconf | netns | nexthop | ntable |
ntbl | route | rule | sr | tap | tcpmetrics |
token | tunnel | tuntap | vrf | xfrm }
OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
-h[uman-readable] | -iec | -j[son] | -p[retty] |
-f[amily] { inet | inet6 | mpls | bridge | link } |
-4 | -6 | -M | -B | -0 |
-l[oops] { maximum-addr-flush-attempts } | -br[ief] |
-o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename]
-rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
-c[olor]}
$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::55f0:9771:cb55:7bf3 prefixlen 64 scopeid 0x20<link>
inet6 fd17:625c:f037:2:eebe:9b47:4b4f:31ac prefixlen 64 scopeid 0x0<g
lobal>
ether 08:00:27:2d:01:bf txqueuelen 1000 (Ethernet)

5. ping:

Verifies IP-level connectivity to a remote host by sending ICMP echo request messages. It is the primary tool for troubleshooting reachability, network latency and name resolution.

* ping google.com.
* ping -c 10 google.com.

RESULT:

Thus the networking commands for windows and linux operating systems.

MATE Terminal

File Edit View Search Terminal Help

```
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<gl
obal>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 119  bytes 33439 (32.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 156  bytes 17151 (16.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 148  bytes 9810 (9.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 148  bytes 9810 (9.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[redacted]$ ping
ping: usage error: Destination address required
[redacted]$ tcpdump
tcpdump: enp0s3: You don't have permission to perform this capture on that devic
e
(socket: Operation not permitted)
```

MATE Terminal

File Edit View Search Terminal Help

```
                -rc[vbuf] [size] | -N[etns]name | -N[umeric] | -a[ll] |
                -c[olor]}
[redacted]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<gl
obal>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 182  bytes 31993 (31.2 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 135  bytes 15261 (14.9 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 48  bytes 3410 (3.3 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 48  bytes 3410 (3.3 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

MATE Terminal

File Edit View Search Terminal Help

```
                                My traceroute  [v0.93]
fedora (::1) -> localhost (::1)                          2020-07-16T11:36:12+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   Quit
                                   Packets               Pings
Host                             Loss%   Snt   Last   Avg  Best  Wrst StDev
1. localhost                      0.0%    30    0.1   0.1   0.0   0.2   0.0
```